20 Two cafe outlets, *A* and *B* sells coffee, tea and cakes.
Both outlets charge \$ 2.50 for a cup of tea, \$3.50 for a cup of coffee and \$3.00 for a piece of cake.

(a) On a Saturday, Cafe *A* sells 100 cups of coffee, 70 cups of tea and 50 pieces of cake.
Cafe *B* sells 120 cups of coffee, 90 cups of tea, and 40 pieces of cake.

Represent this information in a 2×3 matrix $\boldsymbol{Q}$.

Answer Q = $\begin{pmatrix} & & \end{pmatrix}$ [1]

(b) By multiplying *Q* with another matrix, calculate the total amount collected by each of the two outlets respectively.

Answer: Outlet *A*: \$ ________

Outlet *B*: \$ ________ [2]

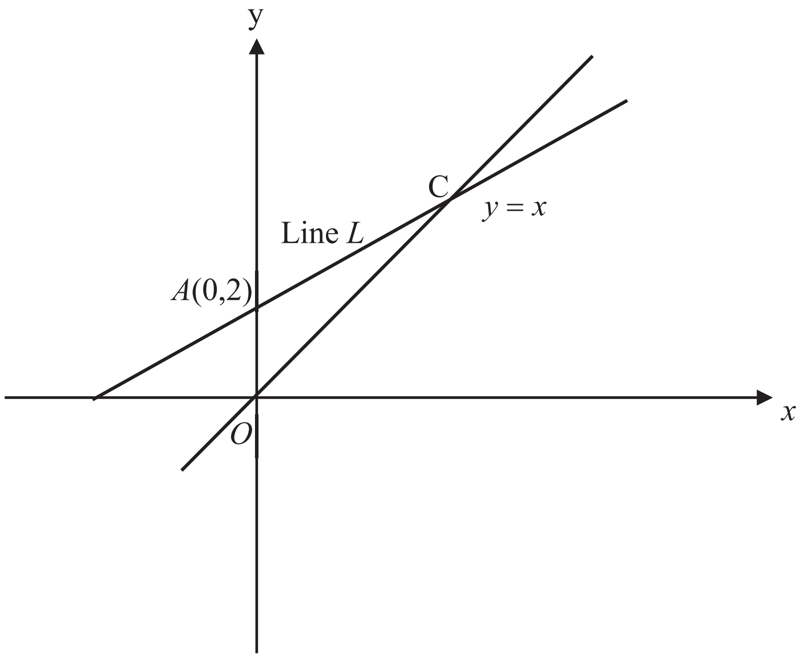
(c) Both outlets decided to give a special discount of 20% for coffee, 10% for tea and 10% for cakes.
By the multiplication of two matrices, calculate the new prices of a cup of coffee, of a cup of tea and of a piece of cake.

Answer: New price of a cup of coffee : \$________,

New price of a cup of tea : \$________,

New price of a piece of cake : \$________[2]

21



The line L passes through the point A (0,2) and meets the line $y = x$ at point C. O is the origin. The area of OAC is 3 units^2.

(a) What is the equation of line L ?

Answer ____________________ [2]

(b) There is a point B such that OC is the line of symmetry of the figure $OACB$.

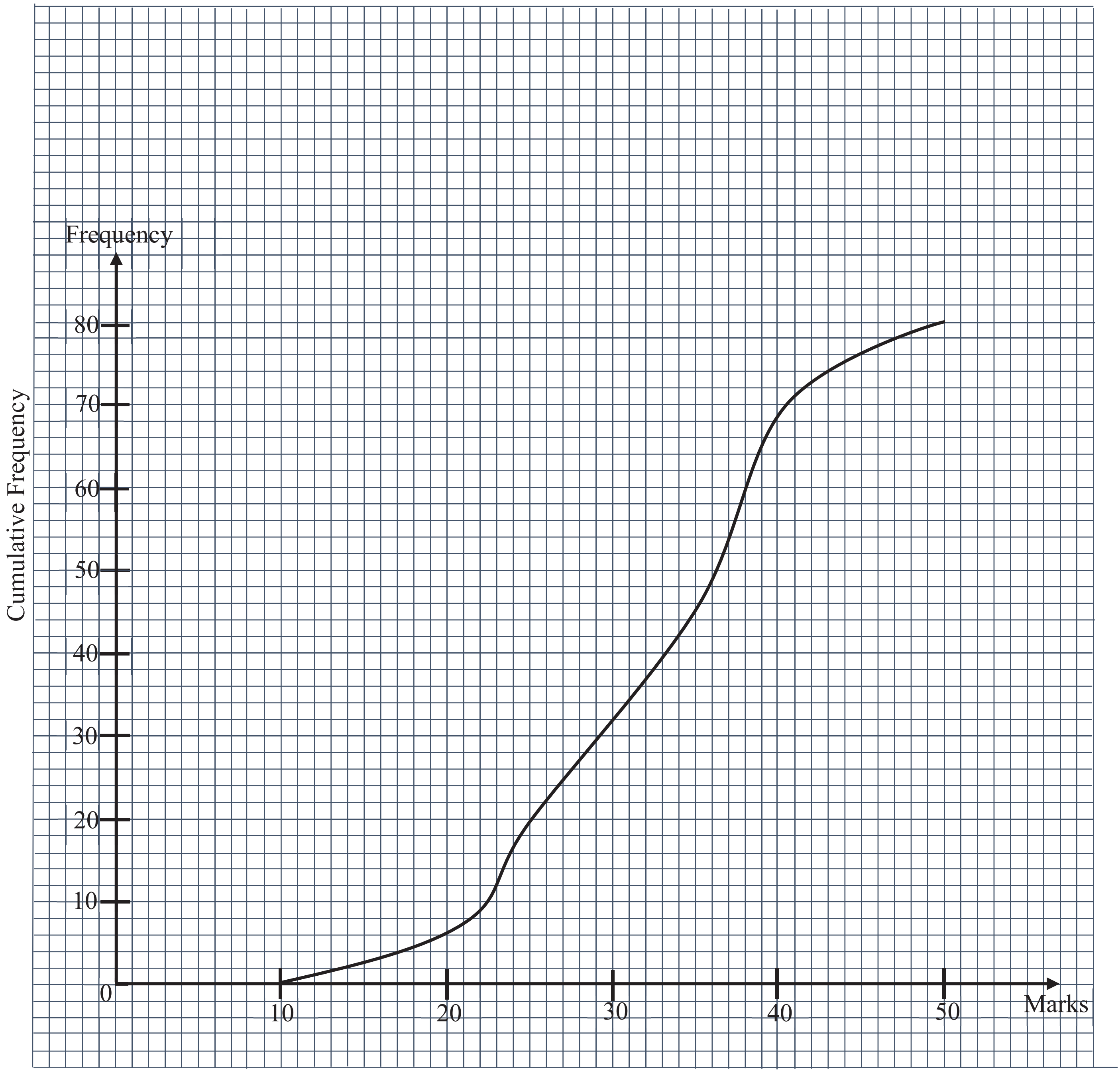
(i) What is the name of quadrilateral $OACB$?

Answer ____________________ [1]

(ii) State the coordinates of the point B.

Answer B (,) [1]

22. The cumulative frequency represents the marks scored by a group of 80 students.



(a) Use the graph to find the number of students with a score of more than 26 marks.

Answer ________________ [1]